Homework Assignment #1

Deliverables

In this assignment, you will:

- Write, debug, and run simple MatLab programs.
- Practice variable assignments and type-casting.
- Perform simple arithmetic and relational computations using MatLab operators.
- Provide simple outputs to the user.

All questions on this assignment are to be completed individually.

Questions 1-3 should be submitted to MATLAB Grader and will be evaluated specifically on the functionality of the code. These questions are equally weighted and amount to a total of 10 points toward the assignment grade. It is recommended that you write and debug your code in MATLAB on your own computer before logging into MATLAB Grader for submission. You have 50 submission attempts on MATLAB Grader.

Questions 4 should be submitted to both Crowdmark and will be evaluated for code functionality and readability (i.e. informative variable names). This question is worth 10 points toward the assignment grade.

To submit to Crowdmark please generate a PDF document for each question, containing:

1. The code that you wrote
2. The command window readout for your code, including input and output statements.
3. Any requested written explanations.

Code files (.m files) for all four problems should be submitted to the LEARN dropbox to aid the grader in interpreting your code and to check for plagiarism. Please ensure that your code files have the following file names:

`hwX_qX_FirstInitialLastName.m` (e.g. `hw1_q4_ALund.m`)

and submit only the code file to the LEARN dropbox. Your submission to the LEARN dropbox should have the same code as your submissions to Crowdmark and MATLAB Grader.

You must submit to both locations to receive full credit. Missing submissions to the LEARN Dropbox will result in a 5% deduction in marks for the assignment. Missing submissions to the Crowdmark site will result in 0 marks for the question.

Question 1

<u>Problem Description</u>

In their preliminary analysis of the site for a future gymnasium, a team of geotechnical engineers has collected moisture content from 6 locations on the property, as shown in Fig. 1. However, when they began their calculations, they realized that their intern had recorded the data using the incorrect label names, as shown in Fig. 2. Help the team reassign their data to the proper labels, so they can complete their calculations.

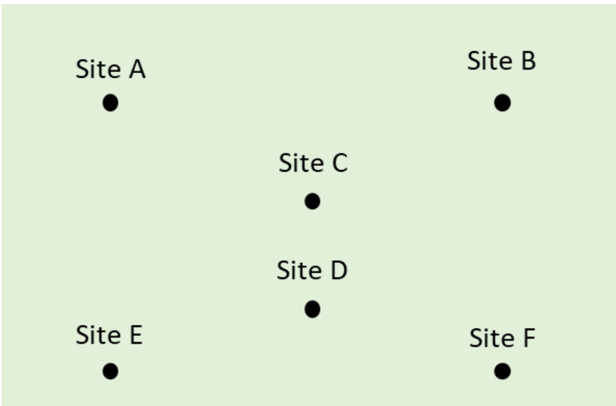

Fig. 1 – Desired Measurement Labels

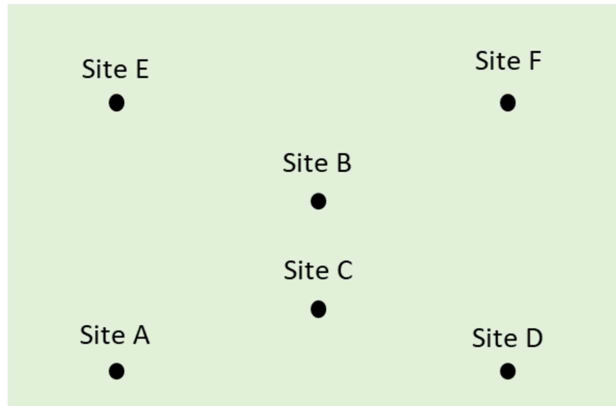

Fig. 2 – Actual Measurement Labels

Note that the output variables must have the same names as the input variables (i.e. moistureA, moistureB, etc.). To generate the original, shuffled data points, please copy and paste the code below.

```
moistureA = rand*60 + 20; % a random number between 20 and 80%
moistureB = rand*60 + 20; % a random number between 20 and 80%
moistureC = rand*60 + 20; % a random number between 20 and 80%
moistureD = rand*60 + 20; % a random number between 20 and 80%
moistureE = rand*60 + 20; % a random number between 20 and 80%
moistureF = rand*60 + 20; % a random number between 20 and 80%
```

Hint: *To check that you've sorted your variables the right way, try printing out moistureA, moistureB, moistureC, moistureD, moistureE, and moistureF at the beginning of your code and end of your code.*

<u>What you need to submit</u>

Copy your completed code to MatLab Grader and press the 'Submit' button to check your work. If the code is correct, no further action is needed. If there are still some bugs, you have the opportunity to resubmit your code for this problem up to 50 times. Your final script file should be submitted to the LEARN Dropbox.

Question 2

Question 2: Logical and Relational Operators in MATLAB

Problem Description

In this question, you will practice using **relational operators** and **logical operators** in MATLAB.

You are given one character variable and three integer variables:

```
ch = '0';   % example character
x = 4;      % example integer
y = 3;      % example integer
z = 2;      % example integer
```

Your task is to create four logical variables:

```
logicalA
logicalB
logicalC
logicalD
```

Each variable should store either `true`/`false` or `1`/`0`.

Tasks

Part A

Check whether the sum of `x`, `y`, and `z` is either:

1. greater than `x^2`, **or**
2. less than `y^3`

Store the result in `logicalA`.

In other words, check:

```
x + y + z > x^2
```

or

```
x + y + z < y^3
```

Part B

Check whether `x`, `y`, and `z` are all different from each other.

Store the result in `logicalB`.

For example:

```
x = 4;
y = 3;
z = 2;
```

All three values are different, so `logicalB` should be `true`.

However:

```
x = 4;
y = 4;
z = 3;
```

Two values are the same, so `logicalB` should be `false`.

Important: Do not write the condition like this:

```
x ~= y ~= z
```

This does not correctly check whether all three values are different in MATLAB.

Instead, compare the variables two at a time.

Part C

Check whether the character stored in `ch` is a number from `'0'` to `'9'`.

Store the result in `logicalC`.

For example:

```
ch = '5';
```

should give:

```
logicalC = true;
```

but:

```
ch = 'm';
```

should give:

```
logicalC = false;
```

You may use the ASCII table for this part.

Part D

Check whether the character stored in `ch` is a **non-alphanumeric character**.

A non-alphanumeric character is a character that is **not**:

- a number from `'0'` to `'9'`
- an uppercase letter from `'A'` to `'Z'`
- a lowercase letter from `'a'` to `'z'`

Store the result in `logicalD`.

For example:

```
ch = '!';
```

should give:

```
logicalD = true;
```

but:

```
ch = 'A';
```

should give:

```
logicalD = false;
```

Example Test Cases

Use the following examples to test your code by hand.

Test Case 1

```
ch = '0';
x = 4;
y = 3;
z = 2;
```

Expected results:

```
logicalA = true
logicalB = true
logicalC = true
logicalD = false
```

Test Case 2

```
ch = 'm';
x = 4;
y = 4;
z = 3;
```

Expected results:

```
logicalA = true
logicalB = false
logicalC = false
logicalD = false
```

Test Case 3

```
ch = '!';
x = 4;
y = 4;
z = 4;
```

Expected results:

```
logicalA = true
logicalB = false
logicalC = false
logicalD = true
```

Important Notes

Relational operators such as <, >, <=, >=, ==, and ~= compare **two values at a time**.

Therefore, avoid writing expressions like:

```
1 < x < 4
```

or

```
x ~= y ~= z
```

These expressions may run in MATLAB, but they do not mean what they look like mathematically.

Instead, break the condition into smaller comparisons and combine them using logical operators such as:

```
&&
```

for **AND**, and

```
||
```

for **OR**.

What You Need to Submit

Submit your completed MATLAB code to MATLAB Grader. Your code should correctly define the following four variables:

```
logicalA
logicalB
logicalC
logicalD
```

Your code should work correctly for different values of `ch`, `x`, `y`, and `z`.

Question 3

<u>Problem Description</u>

The ideal gas law describes the relationship between pressure (P), temperature (T), volume (V), and the number of moles of gas (n).

$$PV = nRT,$$

where R is the ideal gas constant ($0.08314472\ L \cdot bar/K/mol$). The ideal gas law is a good approximation of the behavior of gasses when the pressure is low and the temperature is high. In 1873, Johannes Diderik van der Waals proposed a modified version of the ideal gas law that better models the behavior of real gases over a wider range of temperatures and pressures.

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT,$$

Where a and b represent values characteristic of individual gases.

Given the input variables:

```
P = randi([90 250]); % Random pressure (bar) in the interval [90,250]
V = randi([1 40])/10; % Random volume (L) drawn from the interval [0.1, 4]
n = randi([1 2]); %Random moles of gas drawn from the set [1, 2]
```

Write a code that:

a) Calculates the temperature of water vapor (steam) using both the ideal gas law (`tempIdeal`) and van der Waals' equation (`tempVDW`) in ***units of degrees Celsius***. You may assume that for water, $a = 5.536\ L^2 \cdot bar/mol^2$ and $b = 0.03049\ L/mol$.

a) Report in the logical variable `idealCorrect` if the ideal gas equation is a sufficient representation (within 5 degrees Celsius, inclusive) of the true temperature described by the van der Waals equation. Assign a value of `true` (1) if the two temperatures are sufficiently close.

<u>What you need to submit</u>

Copy your completed code to MatLab Grader and press the 'Submit' button to check your work. If the code is correct, no further action is needed. If there are still some bugs, you have the opportunity to resubmit your code for this problem up to 50 times. Your final script file should be submitted to the LEARN Dropbox.

Question 4

<u>Problem Description</u>

Type each expression into MatLab and report their results to the user in an informative way, such as:

```
fprintf("The output of 1 + 1 is: %d.", outputVar)
```

Then, provide a 1-2 sentence explanation for each expression as to order of operations and class conversions performed by MatLab to achieve the result.

a)

```
2 * -3 ^ 3 + 4
```

b)
```
~(5 ~= 10 / 2)
```

c)
```
2 < 4 < 3
```

d)
```
'A' + 5 - 'B'
```

e)
```
int16((str2num('12') * 3 - 6) / 4)
```

<u>What you need to submit</u>

The code, command window display, and explanations for each expression should be submitted in a single PDF file to Crowdmark. Your script file should be submitted to the LEARN Dropbox.